Homework 06

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In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import roc_curve, roc_auc_score
%matplotlib inline

# Setup — use local data file (avoids macOS SSL certificate issues with URLs
data = pd.read_csv("data/breast-cancer.csv")

predictors = ["mean_radius", "mean_texture", "mean_smoothness"]
X = data[predictors]
y = data["y"]


X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=43206,
    stratify=y
)


scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Question 1: Compare KNN and logistic regression

Question 1

(Your written answer.)

In [2]:
```
logistic_fit = LogisticRegression(
    C=np.inf,
    solver="lbfgs",
    tol=1e-10,
    max_iter=1000
)
logistic_fit.fit(X_train, y_train)
```

```
q1_probabilities = {
    "Logistic regression": logistic_fit.predict_proba(X_test)[:, 1]
}


for k in [5, 15, 25, 35, 45]:
    knn_fit = KNeighborsClassifier(
        n_neighbors=k,
        weights="uniform",
        metric="euclidean"
    )
    knn_fit.fit(X_train, y_train)
    q1_probabilities[f"KNN (k={k})"] = knn_fit.predict_proba(X_test)[:, 1]


q1_results = pd.DataFrame([
    {
        "Model": name,
        "Test AUC": roc_auc_score(y_test, probabilities)
    }
    for name, probabilities in q1_probabilities.items()
])

print("Question 1: Test AUCs")
print(q1_results.to_string(
    index=False,
    formatters={"Test AUC": "{:.4f}".format}
))


plt.figure(figsize=(8, 7))

for name, probabilities in q1_probabilities.items():
    fpr, tpr, _ = roc_curve(y_test, probabilities)
    auc = roc_auc_score(y_test, probabilities)
    plt.plot(fpr, tpr, label=f"{name} (AUC = {auc:.4f})")

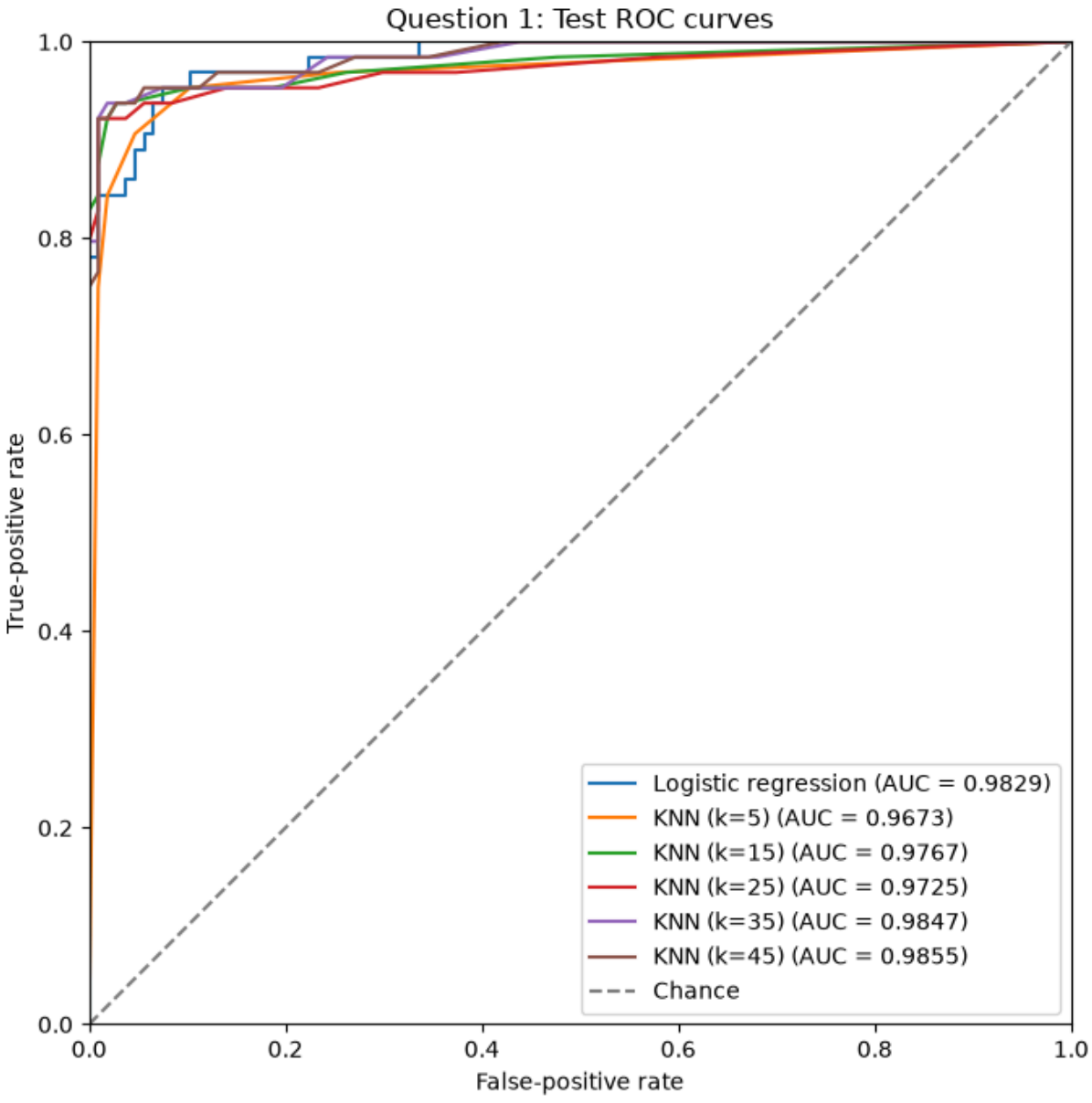
plt.plot([0, 1], [0, 1], "--", color="gray", label="Chance")
plt.xlabel("False-positive rate")
plt.ylabel("True-positive rate")
plt.title("Question 1: Test ROC curves")
plt.xlim(0, 1)
plt.ylim(0, 1)
plt.gca().set_aspect("equal", adjustable="box")
plt.legend(loc="lower right")
plt.tight_layout()
plt.show()



#I would choose KNN with \(k=45\) based on the test AUC, since it
#  had the highest value. It ranked malignant cases above benign
```

```
# cases slightly better than the other models on this split. The difference
# from logistic regression is small, though: its AUC was 0.0026 higher.
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packa
ges/sklearn/linear_model/_logistic.py:1170: UserWarning: Setting penalty=Non
e will ignore the C and l1_ratio parameters
  warnings.warn(
Question 1: Test AUCs
                Model Test AUC
Logistic regression   0.9829
          KNN (k=5)   0.9673
         KNN (k=15)   0.9767
         KNN (k=25)   0.9725
         KNN (k=35)   0.9847
         KNN (k=45)   0.9855
```



Question 2: Training and test ROC curves

Question 2

(Your written answer.)

In [3]:

```
k_values = [1, 3, 5, 10, 20]
q2_probabilities = {}

for k in k_values:
    knn_fit = KNeighborsClassifier(
        n_neighbors=k,
        weights="uniform",
        metric="euclidean"
    )
    knn_fit.fit(X_train, y_train)


    p_train = knn_fit.predict_proba(X_train)[:, 1]
    p_test = knn_fit.predict_proba(X_test)[:, 1]

    q2_probabilities[k] = (p_train, p_test)

q2_results = pd.DataFrame([
    {
        "k": k,
        "Training AUC": roc_auc_score(y_train, p_train),
        "Test AUC": roc_auc_score(y_test, p_test)
    }
    for k, (p_train, p_test) in q2_probabilities.items()
])

print("\nQuestion 2: Training and Test AUCs")
print(q2_results.to_string(
    index=False,
    formatters={
        "Training AUC": "{:.4f}".format,
        "Test AUC": "{:.4f}".format
    }
))

fig, axes = plt.subplots(2, 3, figsize=(13, 8), sharex=True, sharey=True)

for ax, k in zip(axes.flat, k_values):
    p_train, p_test = q2_probabilities[k]

    train_auc = roc_auc_score(y_train, p_train)
    test_auc = roc_auc_score(y_test, p_test)

    fpr_train, tpr_train, _ = roc_curve(y_train, p_train)
    fpr_test, tpr_test, _ = roc_curve(y_test, p_test)

    ax.plot(
        fpr_train,
        tpr_train,
```

```
        label=f"Training (AUC={train_auc:.4f})"
    )
    ax.plot(
        fpr_test,
        tpr_test,
        label=f"Test (AUC={test_auc:.4f})"
    )
    ax.plot([0, 1], [0, 1], "--", color="gray")

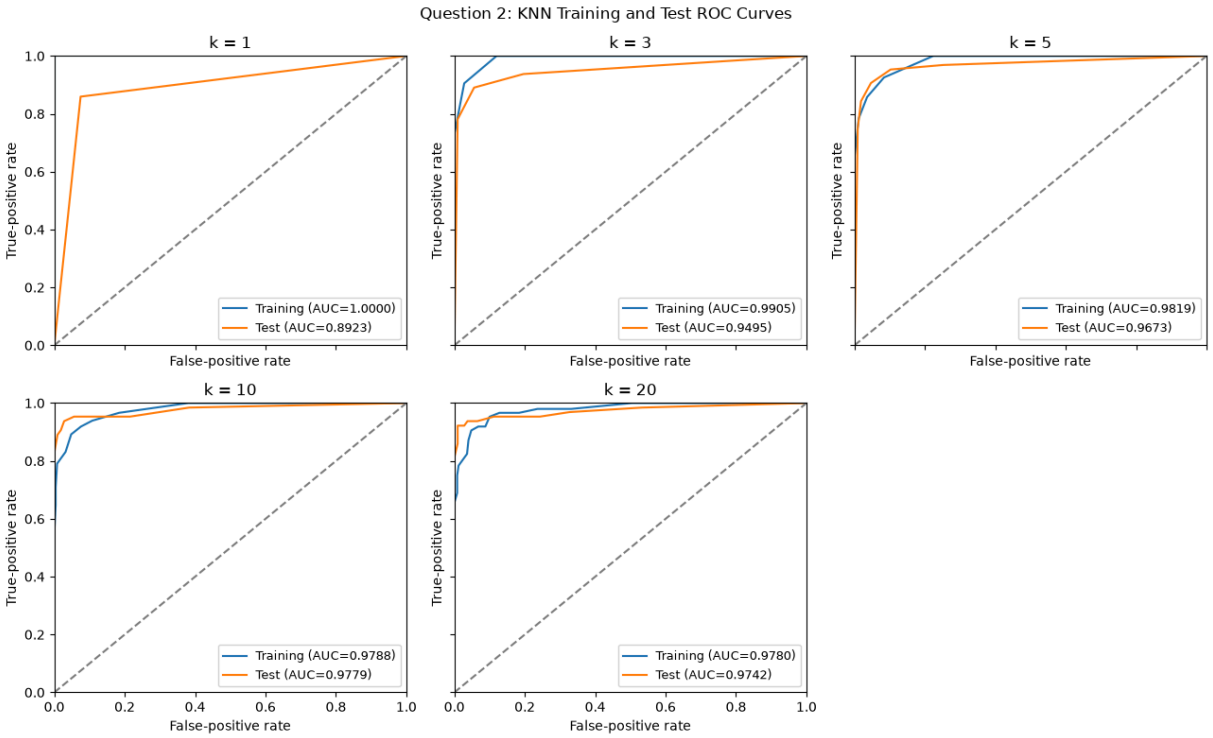
    ax.set_title(f"k = {k}")
    ax.set_xlabel("False-positive rate")
    ax.set_ylabel("True-positive rate")
    ax.set_xlim(0, 1)
    ax.set_ylim(0, 1)
    ax.legend(loc="lower right", fontsize=9)


axes.flat[-1].axis("off")

fig.suptitle("Question 2: KNN Training and Test ROC Curves")
fig.tight_layout()
plt.show()

#At (k=1), each training observation is included among its own nearest neigh
# Its predicted probability is therefore its actual label—0 or 1—so the
#  training AUC is perfect. A new test observation is not its own neighbor,
# and the test AUC is lower.
#As (k) increases, each prediction averages over more neighbors.
# Training AUC decreases, while test AUC improves through \(k=10\) in these
# (k=20). The best test AUC among these choices is at \(k=10\).
# Training performance can overstate performance on new observations
# because the training predictions use observations the model has already se
#The AUCs above came from the earlier run using scikit-learn's bundled
#  copy of the Wisconsin dataset. The code here reads the course CSV directl
#  so its printed AUCs are the ones to use if the CSV's row ordering produce
# different seeded split.
```

```
Question 2: Training and Test AUCs
 k Training AUC Test AUC
 1       1.0000   0.8923
 3       0.9905   0.9495
 5       0.9819   0.9673
10       0.9788   0.9779
20       0.9780   0.9742
```



Question 3: Peer evaluation of Homework 5

Question 3

(Your written answer.)

Report: https://github.com/NeelN86/stat432-fall2026/blob/main/homework/week-05/homework-05.pdf

Issue: https://github.com/NeelN86/stat432-fall2026/issues/2